TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Wafer-Level Acoustic Testing and AI-Based Quality Assessment for Sound-Emitting MEMS Using Probe-Card Integrated Sensors			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Nano/Micro Mechanical Systems	System Characterization & Reliability Evaluation Technologies	Semiconductor Equipment	Measurement/Inspection Equipment
Concept of Proposed Technology	In today's micro-electromechanical systems (MEMS) manufacturing, especially for sound-emitting devices such as loudspeakers, pumps, and ultrasonic actuators, quality control at wafer level is predominantly based on electrical tests [1]. While these tests confirm that contacts and basic electrical functions are intact, they do not reliably reveal mechanical defects such as stiction, incomplete release, contamination, or damaged membranes. As a result, wafers with mechanically defective but electrically "good" MEMS can pass early screening, causing yield loss and additional cost at later stages, particularly after packaging. Optical inspection methods (laser vibrometry, microscopes, confocal sensors) [2] can detect many mechanical issues but are expensive, and depending on the specific technology, unsuitable for high-throughput inline testing. Due to the increasing device density on modern wafers, many optical methods cannot test large numbers of MEMS structures in parallel. Furthermore, some optical techniques are material- and geometry-dependent and become impractical for structures that are			

	partially hidden or already encapsulated. The proposed technology [3] addresses these limitations by introducing acoustic wafer-level testing: MEMS actuators are stimulated electrically via a probe card, and their airborne and/or structure-borne sound is captured by microphones and vibration sensors integrated close to the wafer. Using data-driven and AI-based signal analysis, each device (or groups of devices) is characterized acoustically, enabling a combined electromechanical quality assessment. Parallel stimulation with device-specific test tones allows rapid detection and localization of defective units, even when many MEMS are excited simultaneously. A key element is the integration of acoustic sensors and shielding structures into the probe card or probe-head environment. This allows Korean probe-card SMEs to transform conventional probe cards into “smart acoustic test probes” that simultaneously provide electrical contact and acoustic measurement capability. The approach is designed for compatibility with existing wafer probers, cleanroom requirements, and typical MEMS foundry workflows, enabling scalable, cost-efficient deployment in high-volume production.
Rationale for Joint R&D Project	To make acoustic wafer-level testing a practical and reliable tool for MEMS foundries, the technology must be validated across diverse device types, wafer layouts, and production environments. This requires close collaboration between an institute with deep expertise in acoustic signal processing and AI, and industrial partners experienced in wafer probing, probe-card design, and MEMS manufacturing. Fraunhofer IDMT contributes long-standing experience in acoustic characterization of MEMS and conventional loudspeakers, development of test benches, and AI-based pattern recognition for anomaly detection. Korean SMEs, particularly those manufacturing advanced probe cards for semiconductor and MEMS foundries, provide critical know-how on mechanical design, contact strategies for MEMS actuators, and integration into real wafer-test lines. Their market access and customer feedback ensure that the resulting solution meets stringent productivity, reliability, and cost requirements. The international collaboration enables: • co-design of probe-card-integrated acoustic sensing solutions optimized for MEMS foundries; • validation on wafers from different MEMS processes (e.g., speakers, pumps, ultrasonic devices) under Korean production conditions; • joint development of product-ready probe-card modules tailored to the needs of Korean MEMS manufacturers. This joint approach significantly reduces technical uncertainty for both parties and accelerates commercialization, positioning Korean probe-card SMEs as early market leaders for acoustic MEMS wafer-test solutions.
Key Words	1. Wafer-Level Acoustic Testing 2. MEMS Actuators 3. Probe-Card Integration 4. AI-Based Quality Control

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	The objective of this project is to develop and validate a probe-card-centric wafer-level acoustic testing solution for sound-emitting MEMS devices that can be integrated into standard wafer probers and production lines. The project aims Objective 1. To develop **probe-card-integrated acoustic sensor modules** and shielding concepts that enable robust, high-SNR measurement of sound-emitting MEMS on wafer level. *Objective 2.* To design **AI-based signal processing and classification algorithms** that provide fast and reliable PASS/FAIL decisions, binning, and basic failure pattern identification based on acoustic emission signatures. *Objective 3.* To implement a **complete wafer-test workflow**, including test signal design, data acquisition, and automatic wafer-map generation, that can be deployed with minimal changes to existing MEMS wafer-probe infrastructure and is attractive for Korean probe-card SMEs and their foundry customers.			
R&D Target Specifications	The target performance specifications of this project are as follows: 1. Achieve reliable acoustic PASS/FAIL classification of sound-emitting MEMS on wafer level with at least 98% detection rate for mechanically/acoustically defective but electrically passing devices, compared to post-packaging reference tests. 2. Enable parallel testing of large MEMS arrays, targeting an effective test time of ≤ 1 second per wafer for predefined wafer layouts using multi-tone or coded excitation schemes. 3. Integrate microphones and, where relevant, structure-borne sound sensors into a probe-card or probe-head module that satisfies cleanroom compatibility and standard wafer-prober interface requirements used in Korean MEMS foundries. 4. Provide automatic wafer-map output (incl. binning information) compatible with standard semiconductor data formats, allowing direct integration into manufacturing execution systems (MES) and yield-analysis tools. 5. Demonstrate a cost advantage over purely optical inspection for specified use cases by reducing the need for expensive optical equipment and manual expert evaluation, quantified as at least 30% reduction in test equipment and operational cost for the targeted test stage.			
R&D Plan	**Phase 1: Concept Development and Laboratory Demonstrator (TRL 3 → 4)** In Phase 1, the focus is on defining the core architecture and validating key principles in a controlled laboratory environment. The main questions are how to place and protect acoustic sensors near the wafer, and which excitation and analysis strategies extract the most reliable information about MEMS mechanical integrity. Tasks include:			

- Designing candidate concepts for integrating microphones and/or vibration sensors into or around a probe card, including acoustic shielding structures to suppress environmental noise and undesired structure-borne sound.
- Developing test signals tailored to MEMS actuators (e.g., frequency sweeps, multi-tone sequences, coded signals) that allow both functional verification and fine-grained acoustic characterization.
- Building a laboratory demonstrator with a simplified probe fixture and exchangeable sensor modules to test different layouts and distances to the wafer.
- Recording acoustic data from representative MEMS actuators (e.g., MEMS loudspeakers, micro-pumps, ultrasonic transducers) under controlled conditions, including intentional defects or process variations.
- Implementing baseline signal-processing and AI models (e.g., spectral analysis, feature extraction, supervised classifiers) to demonstrate feasibility of PASS/FAIL classification and simple binning.

The main milestone of this phase is the demonstration of a lab-scale system showing reproducible detection of mechanically/acoustically defective MEMS devices on wafer-level test structures, with clear performance advantages over purely electrical tests for the selected sample set.

Phase 2: Probe-Card Integration and Pilot System for Wafer Probers (TRL 4 → 5)
While Phase 1 focuses on proving the acoustic testing concept, Phase 2 addresses industrial integration, with an emphasis on probe-card design and compatibility with wafer probers used in MEMS foundries.

The Korean SME partner and the Fraunhofer IDMT will:

- Co-design probe-card concepts that integrate microphones, cabling, and local shielding into the limited available space near the contact area, considering thermal, mechanical, and contamination constraints.
- Develop mechanical and electrical interfaces to standard wafer probers, ensuring that the acoustic module can be mounted and exchanged similarly to conventional probe cards.
- Optimize acoustic paths between the MEMS devices and the sensors via the probe card acoustic design to enhance signal-to-noise ratio while maintaining probing performance.
- Refine AI-based classification models, including multi-device testing strategies (e.g., each device emitting a defined frequency or pattern) to exploit spectral “gaps” for defect localization and to allow highly parallel testing.
- Implement end-to-end software for test control, data acquisition, result visualization, and automatic generation of wafer maps with per-die binning information.

Pilot tests will be conducted on wafers supplied either by project partners or collaborating MEMS foundries, using real wafer layouts and realistic throughput requirements. The milestone of this

	phase is a pilot system installed on a wafer prober that performs acoustic wafer-level testing with an integrated probe-card module, achieving the target specifications for detection performance, test time, and operational robustness. **Phase 3: Field Validation, Optimization, and Pre-Commercialization (TRL 5 → 6)** Building on the pilot system, Phase 3 aims to validate the approach in real production or near-production environments at Korean MEMS foundries and to prepare probe-card-based acoustic test modules for commercialization by the SME partner. Activities include: • Deploying the probe-card-integrated acoustic test solution at one or more foundry or Outsourced Semiconductor Assembly and Test (OSAT) lines, operated by or together with Korean SME partners. • Collecting long-term performance data, including device variability across lots, environmental changes (noise, temperature), and equipment drift, and refining models and test procedures accordingly. • Extending the solution to multiple MEMS product types (e.g., loudspeakers, pumps, ultrasonic devices) to demonstrate versatility and broaden the business case for probe-card suppliers. • Validating economic impact by measuring reductions in downstream failures, rework, and reliance on optical inspections, and by quantifying potential yield improvements and binning-based product differentiation. • Finalizing design guidelines and reference implementations for commercial probe cards with integrated acoustic sensing, including documentation, training materials, and support tools for rapid customization for new MEMS designs. The final milestone is the demonstration of a robust, scalable, and cost-effective wafer-level acoustic testing solution, implemented as a probe-card module validated by Korean industrial partners, and ready for productization as an added-value offering by Korean probe-card SMEs.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will lead the core research on acoustic measurement concepts, signal processing, and AI-based classification for wafer-level testing of MEMS actuators. Its primary role is to ensure that the acoustic and algorithmic components achieve the required sensitivity, robustness, and speed for integration into industrial probe-based test systems. Specifically, Fraunhofer IDMT will: 1. Design and validate acoustic test methodologies for MEMS actuators on wafer level, including test signal design and measurement setups. 2. Develop signal-processing and AI algorithms for PASS/FAIL classification, binning, and basic failure-type indication, optimized for short measurement times and noisy industrial

	environments. 3. Investigate parallel testing strategies (e.g., multi-tone excitation, coded excitation) that allow many devices to be tested simultaneously while preserving the ability to localize defects. 4. Support the mechanical and acoustic design of probe-card-integrated sensor modules, including shielding concepts, sensor placement, and vibration isolation from the prober. 5. Conduct systematic experiments, simulations, and performance evaluations to verify detection rates, false-alarm rates, and robustness against process variations. Fraunhofer IDMT will provide laboratory infrastructure, measurement equipment, and expertise in acoustic metrology and AI-based anomaly detection. It will also contribute to intellectual property generation (e.g., patents related to acoustic wafer-level testing concepts) and scientific dissemination.
Expected R&R of Korean SME Partner	The Korean SME partner is envisioned to be a probe-card manufacturer or a company closely collaborating with MEMS probe-card suppliers and foundries. Its role is crucial to ensure that the developed acoustic testing technology can be realized as a viable probe-card product and adopted by MEMS manufacturing customers. Specifically, the SME partner will: 1. Lead the mechanical and electrical design of probe cards that integrate acoustic sensors and associated components, while maintaining or improving probing performance and lifetime. 2. Ensure compatibility with standard wafer probers and MEMS test flows, including handling of special requirements for actuators (e.g., higher currents, specific driving waveforms). 3. Provide practical constraints and design guidelines from the perspective of manufacturing, cost, reliability, and maintenance of probe cards operated in Korean foundries. 4. Host pilot installations and field tests on real production or engineering wafers, in cooperation with MEMS foundries or OSATs in Korea. 5. Prepare commercialization plans, including defining product variants (e.g., generic acoustic test cards, product-specific cards), pricing models, and service concepts for local customers. By taking these roles, the Korean SME partner will transform the project outcomes into differentiated probe-card products and services. This enables them to move beyond conventional electrical-only probe cards and offer integrated electromechanical/electroacoustic test capabilities, strengthening their competitiveness in both domestic and international MEMS markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver both technological innovation and tangible business value for Korean SMEs and the broader MEMS ecosystem. From a technological perspective, the project will establish a new class of wafer-level test methods that combine electrical and acoustic measurements. This enables detection of mechanical defects that are invisible to pure electrical tests, improves understanding of process-induced variations, and supports more precise binning of MEMS actuators according to acoustic performance metrics. The integration of AI methods will ensure that complex acoustic signatures are translated into easy-to-use quality indicators without requiring expert interpretation. For Korean probe-card manufacturers and MEMS-test solution providers, the technology opens up a new value-added product segment: probe cards with integrated acoustic sensing and analysis capability. These “smart” probe cards allow MEMS foundries to add an acoustic quality-control layer without major changes to their wafer probers, thereby increasing yield and reducing downstream scrap and rework. The ability to offer such advanced probe cards early provides a strong differentiation against international competitors. For MEMS foundries and end customers (e.g., smartphone, hearable, medical device, and industrial sensor manufacturers), improved wafer-level screening leads to more consistent product quality, reduced need for post-packaging calibration, and better support for multi-device applications where tight matching between individual MEMS actuators is essential. The resulting efficiency gains and quality improvements support high-volume MEMS manufacturing and enable more ambitious system designs. At a broader industrial and economic level, the collaboration will strengthen the position of Korean SMEs in the semiconductor and MEMS supply chain, foster technology transfer from European research to Korean manufacturing, and contribute to Korea’s strategic goals in semiconductor and AI-driven manufacturing technologies.

6. References

(1) Shoaib, Muhammad, Hamid, Nor Hisham, Malik, Aamir Farooq, Zain Ali, Noohul Basheer, Tariq Jan, Mohammad, A Review on Key Issues and Challenges in Devices Level MEMS Testing, *Journal of Sensors*, 2016, 1639805, 14 pages, 2016. https://doi.org/10.1155/2016/1639805

(2) Zhao, F.; Tang, H.; Zou, X.; Li, X. A Review of Optical Metrology Techniques for Advanced Manufacturing Applications. *Micromachines* **2025**, *16*, 1224. https://doi.org/10.3390/mi16111224

(3) Patent “Methods and apparatuses for acoustically testing MEMS devices”, patent number: EP4097042B1, patent filed: 28th of June 2022

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization